WS07 - States of matter and particle behavior

Name: _____ Date: _____

20–25 minutes total | 20 points | Question page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Objectives: Describe solids, liquids and gases with a particle model; connect structure to shape, volume and compressibility; distinguish a model drawing from actual particle size.

WS07-Q01 (2 points) Describe the particle arrangement and motion in a typical crystalline solid.

WS07-Q02 (2 points) Which pair best describes a liquid under ordinary conditions? A. Fixed shape and widely separated particles. B. Nearly fixed volume and particles that rearrange past close neighbors. C. No volume and no motion. Explain your choice.

WS07 - States of matter and particle behavior

Name: _____ Date: _____

20–25 minutes total | 20 points | Question page 2 of 4

Difficulty: Beginner through application; final question is a challenge

WS07-Q03 (2 points) Why is a gas usually much easier to compress than a liquid?

WS07-Q04 (2 points) A sealed sample changes from solid to liquid. Does its total mass change? Explain without assuming the two phases have equal volume.

WS07 - States of matter and particle behavior

Name: _____ Date: _____

20–25 minutes total | 20 points | Question page 3 of 4

Difficulty: Beginner through application; final question is a challenge

WS07-Q05 (3 points) A hypothetical solid has mass 54.0 g and volume 60.0 mL. Calculate its density to three significant figures.

WS07-Q06 (3 points) Draw the same 10 model particles first as a liquid and then as a gas in equal-size boxes. Add motion arrows and explain one limitation of your drawing.

WS07 - States of matter and particle behavior

Name: _____ Date: _____

20–25 minutes total | 20 points | Question page 4 of 4

Difficulty: Beginner through application; final question is a challenge

WS07-Q07 (3 points) The solid in Q05 melts into 50.0 mL of liquid without losing mass. Calculate the new density and state which phase is denser.

WS07-Q08 (3 points) Challenge: A sealed gas model contains 30 dots in 2.0 L. It expands to 5.0 L. Find the model dot density before and after, then explain why the gas particles have not grown larger.

WS08 - Phase transitions

Name: _____ Date: _____

20–25 minutes total | 20 points | Question page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Objectives: Name and distinguish six phase transitions; identify energy absorbed or released; explain evaporation, boiling and condensation using particles.

WS08-Q01 (2 points) Name solid→liquid and liquid→solid transitions.

WS08-Q02 (2 points) Name gas→liquid and solid→gas transitions.

WS08 - Phase transitions

Name: _____ Date: _____

20–25 minutes total | 20 points | Question page 2 of 4

Difficulty: Beginner through application; final question is a challenge

WS08-Q03 (2 points) For the sample undergoing the transition, which absorbs heat: melting or freezing? State what the other process does.

WS08-Q04 (2 points) A wet towel dries at 24 °C. Must its water have boiled? Name the applicable process.

WS08 - Phase transitions

Name: _____ Date: _____

20–25 minutes total | 20 points | Question page 3 of 4

Difficulty: Beginner through application; final question is a challenge

WS08-Q05 (3 points) A 75.0 g sealed sample of ice is partly melted. There are 27.0 g of liquid water. Find the remaining ice mass and percentage melted.

WS08-Q06 (3 points) Water vapor forms liquid droplets on a cooler surface. Name the transition, identify the energy direction for the vapor becoming liquid, and state whether H_2O becomes a new chemical substance.

WS08 - Phase transitions

Name: _____ Date: _____

20–25 minutes total | 20 points | Question page 4 of 4

Difficulty: Beginner through application; final question is a challenge

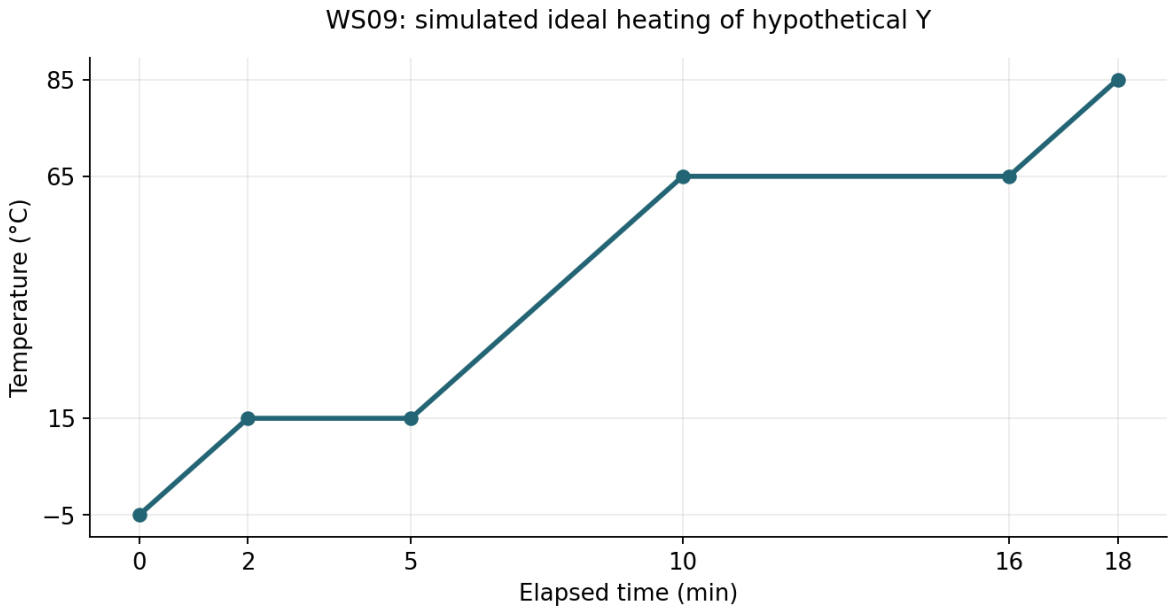
WS08-Q07 (3 points) During one modeled interval, 9.0 g evaporates and 4.0 g condenses inside a closed container. Find the net liquid-mass change and explain what happens to total mass.

WS08-Q08 (3 points) Challenge: A student labels a frost crystal forming directly from water vapor “freezing” and says it absorbs heat. Correct both claims and explain the particle change.

WS09 - Reference page

Keep this page beside the four question pages that follow.

Hypothetical Y, pure substance at constant pressure; ideal equilibrium transitions and straight segments. The sample is solid at the start. Net power into the sample is a constant 80 W. These are simulated teaching data, not a real experiment.



time_min	temperature_C
0	-5
2	15
5	15
10	65
16	65
18	85

WS09 – Heating and cooling curves

Name: _____ Date: _____

20–25 minutes total | 20 points | Question page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS09 reference page included immediately before these question pages.

Objectives: Read axes, slopes and plateaus; distinguish temperature change from phase change; calculate interval energy from stated net power without calculus.

WS09-Q01 (2 points) Use the WS09 reference graph for hypothetical Y. Identify both axes and their units.

WS09-Q02 (2 points) What are the melting and boiling temperatures in the stated Y model?

WS09 - Heating and cooling curves

Name: _____ Date: _____

20–25 minutes total | 20 points | Question page 2 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS09 reference page included immediately before these question pages.

WS09-Q03 (2 points) At t=4 min, name the coexisting phases and state whether heat can still enter.

WS09-Q04 (2 points) Which interval shows warming of liquid only: 0–2, 2–5, 5–10, 10–16, or 16–18 min? Explain.

WS09 - Heating and cooling curves

Name: _____ Date: _____

20–25 minutes total | 20 points | Question page 3 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS09 reference page included immediately before these question pages.

WS09-Q05 (3 points) Find the average temperature-change rate from t=5 min to t=10 min, including units.

WS09-Q06 (3 points) The constant net input is 80 W. Find the energy received during the boiling plateau from 10 to 16 min.

WS09 - Heating and cooling curves

Name: _____ Date: _____

20–25 minutes total | 20 points | Question page 4 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS09 reference page included immediately before these question pages.

WS09-Q07 (3 points) Sketch an ideal cooling curve for the same pure Y at the same pressure, beginning with gas at 85 °C and ending with solid at −5 °C. Label both plateau temperatures and their transitions; exact time lengths are not specified.

WS09-Q08 (3 points) Challenge: During the melting plateau, net power stays constant and all input drives melting. What fraction of the sample has melted at 3 min 30 s? Assume all solid at 2 min and all liquid at 5 min. Explain why using temperature alone cannot find this fraction.

WS10 - Sensible heat

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Objectives: Choose and rearrange Q=mcΔT for a single phase; keep units and signs consistent; separate heat received by water from energy supplied to a whole setup.

WS10-Q01 (2 points) What does c represent in Q=mcΔT? Give one consistent unit for c.

WS10-Q02 (2 points) A sample warms from 12.0 °C to 29.0 °C. Find ΔT and the sign of Q in the single-phase heating model.

WS10 - Sensible heat

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

Difficulty: Beginner through application; final question is a challenge

WS10-Q03 (2 points) Which requires more heat for the same rise: 50 g or 150 g of the same liquid, with the same c? Give the ratio.

WS10-Q04 (2 points) A problem gives m=0.200 kg and c=4.18 J/(g·°C). What mass should be used with that c?

WS10 - Sensible heat

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

Difficulty: Beginner through application; final question is a challenge

WS10-Q05 (3 points) Calculate heat needed to warm 120 g water from 20.0 °C to 35.0 °C. Use c=4.18 J/(g·°C); ignore container effects and losses. Give three significant figures.

WS10-Q06 (3 points) A 50.0 g metal sample with c=0.450 J/(g·°C) cools from 80.0 °C to 30.0 °C. Give signed Q and the amount released.

WS10 - Sensible heat

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

Difficulty: Beginner through application; final question is a challenge

WS10-Q07 (3 points) An 80.0 g water sample starts at 18.0 °C and receives 1672 J. Use c=4.18 J/(g·°C). Find final temperature under the single-phase model.

WS10-Q08 (3 points) Challenge: A device source supplies 12.0 kJ during an illustrative trial. The 100 g water warms by 8.0 °C, with c=4.18 J/(g·°C). Find the modeled heat gained by the water and the percentage of source energy this represents. Explain why the remainder need not violate conservation.

WS11 - Latent heat and multistage changes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Objectives: Use Q=mL for phase transitions; determine partial melting; combine sensible and latent stages without skipping boundaries.

WS11-Q01 (2 points) Which equation models melting at constant temperature: Q=mcΔT or Q=mL? Explain.

WS11-Q02 (2 points) State one valid unit of latent heat L and explain why it has no temperature-change denominator.

WS11 - Latent heat and multistage changes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

Difficulty: Beginner through application; final question is a challenge

WS11-Q03 (2 points) For water condensing at its boiling temperature under fixed pressure, is Q positive or negative for the condensing water? What about the surroundings receiving that heat?

WS11-Q04 (2 points) A 30 g ice sample is already at 0 °C. Is “at 0 °C” enough to conclude it is all liquid? Explain.

WS11 - Latent heat and multistage changes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

Difficulty: Beginner through application; final question is a challenge

WS11-Q05 (3 points) Find the heat required to melt 25.0 g ice at 0 °C to water at 0 °C near 1 atm. Use $L_f=334$ J/g.

WS11-Q06 (3 points) A 60.0 g ice sample at 0 °C receives 6680 J near 1 atm. Use $L_f=334$ J/g and neglect other energy terms. Find liquid mass, remaining ice and final temperature.

WS11 - Latent heat and multistage changes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

Difficulty: Beginner through application; final question is a challenge

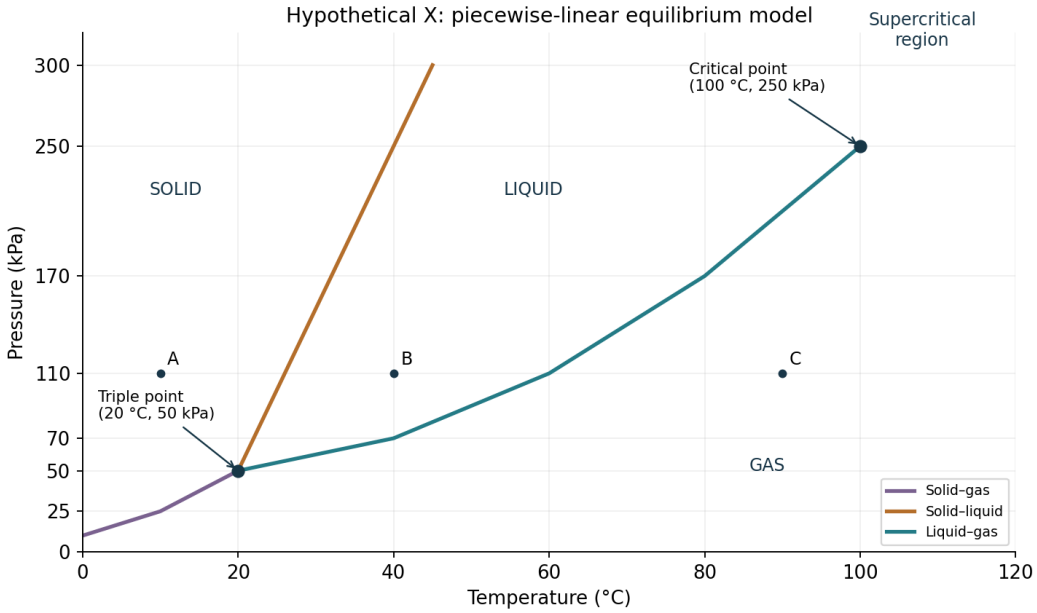
WS11-Q07 (3 points) How much heat is released when 3.00 g water vapor condenses to liquid at 100 °C near 1 atm? Use L_v=2260 J/g. Give the signed heat for the water too.

WS11-Q08 (3 points) Challenge: Heat 15.0 g ice from −8.0 °C to liquid water at 12.0 °C near 1 atm. Use c_ice=2.10 J/(g·°C), L_f=334 J/g and c_water=4.18 J/(g·°C). Neglect losses and container energy. Show all three stages and total heat.

WS12 - Reference page

Keep this page beside the four question pages that follow.

Hypothetical X equilibrium phase diagram; model data, not a measured material. Connect consecutive points on each named boundary by straight segments. Solid-liquid line: P=10T−150 over 20≤T≤45, with kPa and °C. Point A=(10 °C,110 kPa); B=(40 °C,110 kPa); C=(90 °C,110 kPa).



boundary	temperature_C	pressure_kPa
solid–gas	0	10
solid–gas	10	25
solid–gas	20	50
solid–liquid	20	50
solid–liquid	45	300
liquid–gas	20	50
liquid–gas	40	70
liquid–gas	60	110
liquid–gas	80	170
liquid–gas	100	250

WS12 - Reading phase diagrams

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS12 reference page included immediately before these question pages.

Objectives: Identify phase regions and coexistence boundaries; follow constant-pressure and constant-temperature paths; distinguish triple and critical points from an ordinary boiling point.

WS12-Q01 (2 points) In the X phase diagram, identify the horizontal and vertical quantities with units.

WS12-Q02 (2 points) Name the stable phase at A=(10 °C,110 kPa) and at C=(90 °C,110 kPa).

WS12 - Reading phase diagrams

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS12 reference page included immediately before these question pages.

WS12-Q03 (2 points) Give the triple-point coordinates and name the phases that can coexist there.

WS12-Q04 (2 points) What is the critical-point coordinate, and which coexistence boundary ends there?

WS12 - Reading phase diagrams

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS12 reference page included immediately before these question pages.

WS12-Q05 (3 points) At 70 kPa, use the stated solid-liquid line $P=10T-150$ to find the melting temperature. The table gives boiling at 40 °C at this pressure. Find the width of the liquid-only temperature interval.

WS12-Q06 (3 points) At T=80 °C, pressure increases slowly from 100 to 200 kPa. Use the boundary table to name the transition and the pressure at which coexistence is crossed.

WS12 - Reading phase diagrams

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS12 reference page included immediately before these question pages.

WS12-Q07 (3 points) A point lies on the liquid-gas boundary at (40 °C,70 kPa). Can you conclude that exactly half the mass is liquid? Explain what is known and what extra information would be needed.

WS12-Q08 (3 points) Challenge: At 25 kPa, heat X from 0 °C to 30 °C. Use the solid-gas boundary point (10 °C,25 kPa). Identify the transition and explain why no stable liquid interval is crossed.

WS13 - Ideal gas law and units

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Objectives: Explain the particle basis of PV=nRT; choose consistent pressure, volume and temperature units; solve for one unknown and test whether the ideal model is appropriate.

WS13-Q01 (2 points) A gas is at 22.0 °C. Give the unrounded Kelvin conversion used in PV=nRT. Explain why 22.0 is not the correct temperature input.

WS13-Q02 (2 points) Which R is appropriate for P in kPa, V in L, n in mol and T in K? A. 8.314 kPa·L/(mol·K). B. 0.08206 L·atm/(mol·K) without any conversion. Explain.

WS13 - Ideal gas law and units

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

Difficulty: Beginner through application; final question is a challenge

WS13-Q03 (2 points) A gas sample is compressed at constant T and n from 4.0 L to 2.0 L. Predict how its ideal-gas pressure changes and give the factor.

WS13-Q04 (2 points) A gauge reads 30 kPa above local atmospheric pressure of 100 kPa. What pressure belongs in PV=nRT?

WS13 - Ideal gas law and units

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

Difficulty: Beginner through application; final question is a challenge

WS13-Q05 (3 points) An ideal gas has n=0.150 mol, T=320 K and V=4.00 L. Use R=8.314 kPa·L/(mol·K). Find absolute pressure to three significant figures.

WS13-Q06 (3 points) A gas occupies 6.00 L at 100 kPa absolute and 300 K. Use R=8.314 kPa·L/(mol·K). Find n to three significant figures.

WS13 - Ideal gas law and units

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

Difficulty: Beginner through application; final question is a challenge

WS13-Q07 (3 points) A sealed rigid vessel contains ideal gas at 120 kPa absolute and 300 K. It cools to 270 K with no gas lost. Find the final pressure and name the two fixed quantities that justify the shortcut.

WS13-Q08 (3 points) Challenge: A 2.00 g gas sample has molar mass 4.00 g/mol, temperature 27.0 °C and absolute pressure 100 kPa. Use R=8.314 kPa·L/(mol·K). Calculate volume and name one condition that could make the ideal model unreliable.